2D Embeddings for Length Generalization in Character-Based Transformers (CBT)

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Abstract

Small autoregressive Transformers fail on multi-digit addition beyond the sequence lengths seen during training, with standard 1D positional baselines collapsing to 0.0% accuracy on unseen lengths. We find an effective solution in two-dimensional positional coordinates (**Decoupled 2D-RoPE**), which preserve the place-value structure of digits invariant to problem length and eliminate positional phase drift. Hypothesizing that this structural principle extends to text and discourse, we apply 2D coordinates directly to atomic characters and multi-turn discourse tags rather than standard subword tokens. Combining character-level 2D coordinates, coordinate shift masking, and weight-tied recurrence into an 18.95M-parameter model (**CBT-18.95M**), we achieve strong algorithmic and tool-routing results at small scale: solving 100% of seen 1–15 digit sums, generalizing zero-shot to 98.8% (237/240) on unseen 16–50 digit random multi-carry additions, and maintaining 100% (26/26) structured tool dispatch within a compact 36 MB edge footprint.

1 Introduction

A foundational question in machine learning is whether autoregressive neural networks can internalize and execute exact sequential algorithms invariant to sequence length. The prevailing paradigm in contemporary language modeling scales autoregressive Transformers [1, 4] to tens or hundreds of billions of parameters over subword Byte-Pair Encoding (BPE) vocabularies. While these models exhibit remarkable fluency on conversational tasks, small Transformers trained from scratch notoriously fail on basic algorithmic tasks like multi-digit addition when tested beyond the sequence lengths seen during training [5, 6, 7, 8, 13, 14]. Frontier models with chain-of-thought or tool access do considerably better; our statements concern small models trained from scratch. While such a Transformer can memorize or interpolate addition patterns up to its training ceiling N_{train}, its accuracy collapses to 0.0% as the number of digits N exceeds that horizon.

1.1 Algorithmic Arithmetic and the Success of 2D Embeddings

Multi-digit integer addition serves as the canonical testbed for algorithmic length generalization: it possesses an unbounded length axis and is governed by a compact, deterministic finite-state machine (FSM) operating right-to-left from the units column ($i = 0$) to the most significant column ($i = N - 1$):

$$s_i = (a_i + b_i + c_{i-1}) \bmod 10, \quad c_i = \left\lfloor \frac{a_i + b_i + c_{i-1}}{10} \right\rfloor \quad (1)$$

where $a_i, b_i \in \{0, \ldots, 9\}$ are the single-digit operands at column i, $c_i \in \{0, 1\}$ is the binary carry bit emitted to column $i+1$, and $c_{-1} = 0$. Because the entire transition table consists of only $10 \times 10 \times 2 = 200$ discrete rules, executing addition requires $\mathcal{O}(N)$ identical state transitions regardless of whether $N = 4$, $N = 15$, or $N = 50$.

When we trained a standard 18.91M Transformer ($d_{\text{model}} = 512, L = 6, n_{\text{head}} = 8$) with standard one-dimensional rotary embeddings (1D-RoPE [2, 14]) on 1- to 12-digit additions with explicit column scratchpads [9, 10], it failed completely beyond its training horizon: achieving 98.4% on seen 1–12 digits and 4.6% on moderate 14–16 digits, before falling to **0.0% accuracy on unseen 18- to 20-digit sums** (Table 8). Diagnosing these runs revealed that 1D positional encodings assign a flat, monotonically increasing index $t = 0, \ldots, T - 1$. At unseen sequence lengths ($T \approx 600$ characters for 50 digits), relative distances $(m - n)$ between an answer column and its operand digits take values never encountered during training, inducing positional phase drift and repetitive zero loops where the model repeats blocks like `(0+0+0)=00` indefinitely:

$$P\big(\texttt{"(0+0+0)=00"} \mid \texttt{"(0+0+0)=00"}^{\times m}\big) \approx 1 \quad (2)$$

When alternative 1D architectures were evaluated under identical budgets (Looped Transformers [11, 12], Hierarchical Reasoning Models, Gated Linear Recurrent Units, Harmonic Oscillators, and discrete Abacus place embeddings [13]), all nine baselines fell to 0.0% on 18–20 digits (Table 8).

To overcome this failure, we structured the problem onto a **two-dimensional (2D) coordinate grid** $(p_t, \tau_t) \in \mathbb{N}^2$, where macro row p tracks decimal place-

value rank (10^p) and micro column $\tau \in [0, 9]$ tracks the intra-column execution step (Table 1A). Using **Decoupled 2D Rotary Position Embeddings (2D-RoPE)**, matching digits across operands, scratchpad, and answer share identical macro coordinates ($\Delta p = 0$). We found that **2D embeddings perform exceptionally well on arithmetic**: by preserving the decimal place-value structure of digits invariant to problem length, 2D coordinates eliminate 1D phase drift and enable reliable length extrapolation far beyond the training boundary (e.g., our 1.56M HRM-RoPE baseline extrapolates to 20 digits using Decoupled 2D-RoPE alone; Table 3).

1.2 From 2D Digits to 2D Characters: The Core Hypothesis

Having found that 2D embeddings succeed at arithmetic because they prevent losing the internal place-value structure of digits across sequence length, we formed a central hypothesis: *Could the exact same structural principle be applied to atomic characters and natural text/tool discourse?*

Standard language models rely on subword BPE tokenization, which clumps arbitrary character and digit substrings into irregular multi-character tokens:

$$\texttt{"12345678"} \xrightarrow{\text{BPE}} \big[\texttt{"12"}, \texttt{"3"}, \texttt{"45"}, \texttt{"67"}, \texttt{"8"}\big]$$

$$\texttt{"87654321"} \xrightarrow{\text{BPE}} \big[\texttt{"8"}, \texttt{"7"}, \texttt{"65"}, \texttt{"43"}, \texttt{"21"}\big]$$

A single token like `"45"` fuses the 10^3 place (4) and the 10^2 place (5) into one vector at one 1D index, breaking 1-to-1 place-value alignment between operands (Table 7). More broadly, subword merges obscure morphological and entity boundaries in natural language. We hypothesized that if we:

1. decompose inputs into **atomic characters** paired with 14 structural discourse tags (e.g., `<USER>`, `<THINK>`, `<CALL>`, `<RESULT>`, `<ANSWER>`), and
2. place these characters onto a **2D discourse matrix** (Table 1B), where macro row $p \in \{0..4\}$ indexes the functional discourse turn and micro column τ tracks character position within that turn,

then 2D coordinates would preserve the fine-grained structural integrity of words, entity boundaries, and tool syntax without positional degradation, just as they preserve decimal place-value structure in multi-digit sums.

1.3 Hypothesizing the 18M CBT Architecture

To test this hypothesis and achieve maximal algorithmic and tool performance within a compact edge footprint ($<$ 20M parameters, $\sim$36 MB FP16 weights), we hypothesized the unified **Character-Based Transformer (CBT-18.95M)** architecture by combining four complementary mechanisms:

1. **Atomic Character Tokenization** ($V = 115$): Preserves exact character boundaries, eliminating subword tokenization chunking and enabling faithful 1-hop pointer copying for tool arguments.
2. **Decoupled 2D-RoPE**: Rotates half of each attention head with macro row p and the other half with micro column τ, anchoring cross-attention invariant to sequence length in both arithmetic and discourse modes.
3. **Coordinate Shift Masking** ($k = 32$): Randomly shifts macro coordinates during training, exposing rotary frequencies to an extended coordinate horizon ($p_{\max} = N_{\text{train}} + k = 15 + 32 = 47$) to enable zero-shot extrapolation up to 50 digits.
4. **Weight-Tied Recurrent Depth** ($R = 3$): Unrolls $L = 6$ physical layers across $R = 3$ recurrent passes, effectively tripling depth to 18 layers without adding weights, providing the non-linear computational capacity needed for multi-step tool reasoning and schema dispatch within a 36 MB footprint.

1.4 Scope, Limitations, and Utility on Low-Resource Devices

Why study in-model addition when a calculator tool exists? Arithmetic here is a testbed for length generalization: it has a compact, exactly checkable algorithm and an unbounded length axis. We do not argue that an in-model adder is a practical substitute for a calculator; the tool-calling results are a separate experiment.

Limitations: (1) The coordinates (p, τ) are hand-designed and assigned by a rule-based parser, not learned; the length-generalization result is therefore conditional on correct coordinate assignment (Section 3.3) and demonstrates column-aligned addition with a fixed-template scratchpad rather than general algorithmic induction. (2) Tool calling is bound to a small set of templates that overlap with training data, the OOD set contains only 18 prompts, near-exact copies are functionally failures, and we do not compare against a fine-tuned small off-the-shelf model. (3) Tool and 12d-ablation test sets are small (26 and 18 for tools, 14 for Table 9; for arithmetic we supplement our 20 structured cases with a 315-problem random sweep on CBT-18.95M in Table 4 and a 12,800-problem random sweep across four equalized 1–15d, $T = 1024$ models in Table 10), every configuration is a single training run, and we report no seed variance. (4) In Table 8, the 18.91M 1D baselines were trained on 1–12 digits without position-offset augmentation (to address this confound directly, Table 10 trains equalized 100k 1D-RoPE, 1D-Offset, Ruoss et al. randomized-position, and 2D-RoPE models on

Table 1: How the Character-Based Transformer (CBT) Organizes Characters as a 2D Matrix (p, τ): Unlike 1D indices $t \in \{0, 1, \ldots, T-1\}$ that grow along a flat line, CBT places atomic characters onto a structured 2D coordinate grid. **Top (Arithmetic Mode)**: Macro row p tracks the decimal column rank (10^p) so matching digits in the prompt, scratchpad, and answer share identical p ($\Delta p = 0$), while micro column $\tau \in [0, 9]$ tracks the 10-step column template. **Bottom (Discourse Mode)**: Macro row $p \in \{0..4\}$ tracks the functional discourse block, while micro column τ tracks the character index within that block.

A. Arithmetic 2D Matrix Mode: Prompt `47+85=` followed by Scratchpad `<THINK>(0+7+5)=12(1+4+8)=13</THINK>=132`

Region	**Prompt Operands** ($A + B =$)							**Scratchpad Column 0** (10^0 **Ones Place; Col 1 uses** $p = 1, \tau = 0..9$)										**Final Sum**				
Token x_t	4	7	+	8	5	=	<THINK>	(	0	+	7	+	5	)	=	1	2	</THINK>	=	1	3	2
1D Index t	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	28	29	30	31	32
Macro p_t **(Row)**	**1**	**0**	0	**1**	**0**	0	0	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	0	0	**1**	**1**	**0**
Micro τ_t **(Col)**	0	0	0	0	0	0	0	0	1	2	3	4	5	6	7	8	9	0	0	0	0	0

B. Natural Language Discourse 2D Matrix Mode: Multi-Turn Tool-Calling Dialogue (`<USER>` → `<THINK>` → `<CALL>` → `<RESULT>` → `<ANSWER>`)

Discourse Row	**Row** $p = 0$**: User**					**Row** $p = 1$**: Thought**		**Row** $p = 2$**: Tool Call**						**Row** $p = 3$**: Result**		**Row** $p = 4$**: Answer**				
Token x_t	<USER>	W	h	a	t...	<THINK>	I	w...		<CALL>	s	e	a	r	c...	<RESULT>	{	"...	<ANSWER>	I	t	...
Macro p_t **(Row)**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**2**	**2**	**2**	**2**	**2**	**2**	**3**	**3**	**3**	**4**	**4**	**4**	**4**
Micro τ_t **(Col)**	0	1	2	3	4	0	1	2	3	0	1	2	3	4	5	0	1	2	0	1	2	3

identical 1–15 digit data with $T = 1024$), several Table 8 baselines are strawmen for arithmetic (e.g., naive BPE without digit splitting), and our Abacus baseline (0.0% in-distribution) reflects an under-tuned implementation rather than the method of McLeish et al. [13]. (5) The only algorithmic task is addition; we do not test tasks where coordinates cannot be assigned by a simple parser (such as multiplication or sorting). (6) On a laptop CPU (Intel Core Ultra 5 125H, FP32), batched KV-cached inference runs at 141–394 tokens/sec (1.1–5.3 tokens/sec uncached single-stream), and we do not benchmark mobile ARM hardware or INT8 quantization. (7) Recurrent depth showed no benefit in our addition ablation (Table 9), and the 1.56M HRM-RoPE model (Table 3) also extrapolates to 20 digits using Decoupled 2D-RoPE alone, confirming that the 2D coordinate mapping—not recurrence or the multi-stage curriculum—is the primary driver of addition extrapolation.

Practical Utility on Low-Resource Devices: We make an explicit practical claim regarding scale. Scaling CBT as-is to frontier LLM sizes would not make practical sense: because CBT generates text *character-by-character* and evaluates $R = 3$ recurrent passes per character, its autoregressive inference latency is several times slower per word than subword BPE models. CBT's primary utility is in **low-resource local inference**: learning intent routing, structured JSON/API tool calling, and exact arithmetic within a tiny 100k to 18.95M parameter footprint suitable for on-device coprocessors. The 36 MB figure reflects the FP16 weight size only: with $R = 3$, per-token compute is roughly that of a ∼57M-parameter dense model and the KV cache grows with character sequence length. On a laptop CPU (Intel Core Ultra 5 125H, FP32), batched KV-cached decoding achieves 141–394 character tokens/sec across 16–100 digit traces (and 1.1–5.3 tokens/sec uncached single-stream); we have not benchmarked mobile ARM hardware or INT8 quantization.

2 Empirical Results

We first present the empirical evaluation of the final converged **CBT-18.95M** checkpoint (`champion_final.pt`, trained across Stage 1 Pretraining, Stage 2 Search Copy SFT, and Stage 3 All-Tools 1-Hop Annealing SFT), demonstrating decent, robust performance at small scale across both long-horizon arithmetic extrapolation and structured tool agency. We then dissect its performance against tokenization ablations, 1D baseline comparisons, and isolated 100k-parameter component ablations. All benchmark evaluations throughout this paper use deterministic greedy decoding (temperature = 0.0, arg max). Across all tables, we explicitly label the **digit ranges seen during training** ($1 \leq N \leq 15$ for CBT-18.95M; $1 \leq N \leq 12$ for baselines) versus **unseen zero-shot extrapolation ranges** ($16 \leq N \leq 100$).

2.1 Unified Capabilities of CBT-18.95M: Sums, Multi-Tool Routing, and OOD Copying

Tables 2, 3, 4, 5, and 6 present the comprehensive evaluation of our final **CBT-18.95M** model (`champion_final.pt`):

1. **Zero-Shot Extrapolation from 15 Seen Digits Across 16–100 Unseen Digits (Tables 3 and 4)**: Trained strictly on numbers of $1 \leq N \leq 15$ digits with shift masking $k = 32$ (exposing macro column coordinates up to $p_{\max} = 15 + 32 = 47$), CBT-18.95M solves **20/20 (100.0%; CI** [83.9, 100.0]%**)** structured 16–50 digit sums (including $\underbrace{99\ldots99}_{50} + 1 = 10^{50}$). To test beyond structured patterns and map the exact breakdown horizon, we evaluated 315 stratified random multi-carry problems across 13 lengths from 16 to 100 digits (Table 4; 70% uniform random digits, 15% dense carries with $a_i + b_i \geq 9$, 15% cascade chains). Across all 6 tested lengths within the shift-masked co-

Table 2: Executive Benchmark Summary of Final Converged CBT-18.95M (`champion_final.pt`): Scores are raw accuracies with 95% Wilson confidence intervals (no pass/fail threshold is applied). Arithmetic extrapolation combines the 20-case structured suite (Table 3) and the 315-problem random multi-carry sweep across 16–100 digits (Table 4); each model was trained once. The OOD copy rows are computed over the 16 correctly routed prompts of Table 6; exact match is the headline metric.

Capability Domain	Benchmark Suite / Evaluation Metric	Scope / Test Cases	Training Exposure	Final Score	95% Wilson CI
1. Standalone Sums (2D Math)	Tier 1: In-Distribution Column Addition	1–15 Digits	**Seen in Training (1–15d)**	**100.0%**	–
	Tier 2: Structured Extrapolation	16–50 Digits ($n = 20$, incl. $99 \ldots 9 + 1$)	**Unseen OOD (+1 to +35d)**	**100.0% (20/20)**	[83.9, 100.0]%
	Tier 3: Random Multi-Carry Extrapolation (Within $p_{\max} = 47$)	16, 20, 25, 30, 35, 40 Digits ($n = 180$)	**Unseen OOD (+1 to +25d)**	**100.0% (180/180)**	[97.9, 100.0]%
	Tier 4: Random Multi-Carry at Horizon ($p_{\max} = 47$)	45 & 50 Digits ($n = 60$; 58/60 Think OK)	**Unseen OOD (+30 to +35d)**	**95.0% (57/60)**	[86.3, 98.3]%
	Pooled Random Multi-Carry (16–50 Digits)	*8 Lengths × 30 Cases ($n = 240$; 238/240 Think)*	***Unseen OOD (+1 to +35d)***	***98.8% (237/240)***	*[96.4, 99.6]%*
	Tier 5: Beyond Shift-Masking Horizon ($p > 47$)	55d (90%), 60d (45%), 70d (6.7%), 80–100d (0%)	**Unseen OOD (+40 to +85d, $p > p_{\max}$)**	**See Table 4**	–
2. Multi-Tool & Search Suite	Multi-Domain Structured Tool Benchmark	50 Multi-Turn Dialogues	Seen Tool Schemas (6 Domains)	**100.0% (50/50)**	[92.9, 100.0]%
	All-Tools 1-Hop Routing Suite (9 Schemas)	26 Phrasing & Edge Cases	Varied Prepositions & Nonces	**100.0% (26/26)**	[87.1, 100.0]%
	Multi-Tool Exact Argument & Entity Extraction	26 Phrasing & Edge Cases	1-Hop Pointer Copy from `<USER>`	**96.2% (25/26)**	[81.1, 99.3]%
3. Zero-Shot OOD Queries	OOD Syntactic Tag Completion (`<THINK>...<CALL>`)	18 Unseen Domain Prompts	**100% Unseen Prompts/Domains**	**100.0% (18/18)**	[82.4, 100.0]%
	OOD Zero-Shot Tool Selection Accuracy	18 Unseen Domain Prompts	**Unseen Science/Film/Cities/Units**	**88.9% (16/18)**	[67.2, 96.9]%
	OOD 1-Hop Character Pointer Copy, **exact match (headline)**	16 Correctly Routed Cases	**Unseen Target Entities**	**50.0% (8/16)**	[28.0, 72.0]%
	secondary: exact or near-exact (≥85% chars)	16 Correctly Routed Cases	Unseen Target Entities	68.8% (11/16)	[44.4, 85.8]%

Table 3: Itemized Arithmetic Length Extrapolation Benchmark (1 to 50 Digits), Separating Seen Training Lengths from Unseen Zero-Shot Extrapolation Lengths: Standard 1D-Char and 50k-BPE models (18.91M) and Dual-Loop HRM-RoPE (1.56M) were trained on **1–12 digits** with a 512-token context window; our **CBT-18.95M** was trained on **1–15 digits** with a 1024-token context window. Part II (16–20 digits) is the clean comparison within the 512-token window: HRM-RoPE (1.56M) extrapolates to 20 digits using Decoupled 2D-RoPE without CBT's three-stage curriculum, whereas the 1D models fail. In Part III (25–50 digits), extending the 18.91M 1D-Char model's RoPE frequency table to $T = 1024$ at inference still yields `0.0% [FAIL at T=1024]`; for a strictly equalized 1–15 digit, $T = 1024$ training comparison (including Ruoss et al. randomized positions), see Table 10.

Digit Length	Training Exposure Status	Representative Test Problem	Expected Output	18.91M 1D-Char	18.91M 50k-BPE	1.56M HRM-RoPE	CBT-18.95M (Ours)
Part I: Seen Training Distribution ($1 \le N \le$ 12–15 Digits — Seen During Training)							
2 Digits	Seen in Training	`15+25=`	40	40 [PASS]	40 [PASS]	40 [PASS]	40 [PASS]
4 Digits	Seen in Training	`1234+5678=`	6912	6912 [PASS]	6912 [PASS]	6912 [PASS]	6912 [PASS]
8 Digits	Seen in Training	`12345678+87654321=`	99999999	99999999 [PASS]	99999999 [PASS]	99999999 [PASS]	99999999 [PASS]
8 Digits	Seen in Training (Zeros)	`10000000+10000000=`	20000000	20000000 [PASS]	20000000 [PASS]	20000000 [PASS]	20000000 [PASS]
12 Digits	Seen in Training	`123456789012+876543210987=`	999999999999	999999999999 [PASS]	999999999999 [PASS]	999999999999 [PASS]	999999999999 [PASS]
12 Digits	Seen in Training (Overflow)	`500000000000+500000000000=`	1000000000000	1000000000000 [PASS]	1000000000000 [PASS]	1000000000000 [PASS]	1000000000000 [PASS]
14 Digits	Unseen for 12d / Seen for 15d	`12345678901234+87654321098765=`	99999999999999	Zero-Loop [FAIL]	Hallucinates [FAIL]	99999999999999 [PASS]	99999999999999 [PASS]
14 Digits	Unseen for 12d / Seen for 15d	`50000000000000+50000000000000=`	10^{14} (15 digits)	Zero-Loop [FAIL]	Hallucinates [FAIL]	6×10^{13} [FAIL]	10^{14} [PASS]
15 Digits	Max Seen Bound (CBT-18.95M)	`500000000000000+500000000000000=`	10^{15} (16 digits)	Zero-Loop [FAIL]	Hallucinates [FAIL]	Untested	10^{15} [PASS]
Part II: Unseen Moderate & Extreme Extrapolation ($16 \le N \le 20$ Digits — Clean Comparison Within 512-Token Context)							
16 Digits	**Unseen OOD (+1d / +4d)**	`1111222233334444+8888777766665555=`	$16 \times$ '9'	Zero-Loop [FAIL]	Hallucinates [FAIL]	$16 \times$ '9' [PASS]	$16 \times$ '9' [PASS]
18 Digits	**Unseen OOD (+3d / +6d)**	$10^{17} + 10^{17} =$	2×10^{17}	Loops 000... [FAIL]	Invalid CoT [FAIL]	2×10^{17} [PASS]	2×10^{17} [PASS]
20 Digits	**Unseen OOD (+5d / +8d)**	`1234...90 + 8765...09=` (20d)	$20 \times$ '9'	Drift [FAIL]	Drift [FAIL]	$20 \times$ '9' [PASS]	$20 \times$ '9' [PASS]
20 Digits	**Unseen OOD (+5d / +8d)**	$10^{19} + 10^{19} =$	2×10^{19}	Loops 000... [FAIL]	Drift [FAIL]	2×10^{19} [PASS]	2×10^{19} [PASS]
20 Digits	**Unseen OOD (+5d / +8d)**	$\underbrace{999\ldots999}_{20} + 1 =$	10^{20} (21 digits)	Loops 000... [FAIL]	Drift [FAIL]	10^{20} [PASS]	10^{20} [PASS]
Part III: Unseen Ultra-Long Horizon Extrapolation ($25 \le N \le 50$ Digits — Exceeds 512-Token Training Context of 12d Models)							
25 Digits	**Unseen OOD (+10 Extra Digits)**	$\underbrace{999\ldots999}_{25} + 1 =$	10^{25} (26 digits)	Drift [FAIL at T=1024]	N/A (Context Limit)	N/A (Context Limit)	10^{25} [PASS]
35 Digits	**Unseen OOD (+20 Extra Digits)**	$\underbrace{111\ldots111}_{35} + \underbrace{888\ldots888}_{35} =$	$35 \times$ '9'	Drift [FAIL at T=1024]	N/A (Context Limit)	N/A (Context Limit)	$35 \times$ '9' [PASS]
50 Digits	**Unseen OOD (+35 Extra Digits)**	`123...890 + 876...109=` (50d)	$50 \times$ '9'	Drift [FAIL at T=1024]	N/A (Context Limit)	N/A (Context Limit)	$50 \times$ '9' [PASS]
50 Digits	**Unseen OOD (+35 Extra Digits)**	$\underbrace{999\ldots999}_{50} + 1 =$	10^{50} (51 digits)	Drift [FAIL at T=1024]	N/A (Context Limit)	N/A (Context Limit)	10^{50} [PASS]

ordinate support ($d \in \{16, 20, 25, 30, 35, 40\}$, $p \le 40 \le p_{\max}$), CBT-18.95M achieves **180/180 (100.0%; 95% Wilson CI** $[97.9, 100.0]\%$) exact match. Across the full 16–50 digit random suite ($n = 240$), it achieves **237/240 (98.8%; CI** $[96.4, 99.6]\%$) final answer exact match and **238/240 (99.2%; CI** $[97.0, 99.8]\%$) `<THINK>` scratchpad exact match, with **zero (delimiter errors** and zero off-by-one readout errors (the only 3 errors at $d = 45, 50$ were single-digit substitutions at the $p \approx 47$ boundary). Beyond $p_{\max} = 47$, accuracy remains **90.0% (18/20)** at 55 digits ($p \le 55$, +8 columns beyond shifted training support), then drops to **45.0% (9/20)** at 60 digits, **6.7% (1/15)** at 70 digits, and **0.0%** at 80–100 digits. This empirically demonstrates that Decoupled 2D-RoPE generalizes reliably across unseen sequence lengths $N > N_{\text{train}}$ *up to the maximum coordinate index* $p_{\max} = N_{train} + k$ *covered by shift masking during training*, and degrades once p enters rotary angles never observed during training.

2. **Stage 3 All-Tools 1-Hop Alignment (Table 5, 100.0% Routing, 96.2% Exact Args)**: In Stage 2 (`step_7000`), applying 1-hop copy induction only to `search` caused prompts with varied prepositions (like `"What is the weather on Rome"` or `"How many miles is 25 km"`) to over-route to `search`. In Stage 3 (`champion_final.pt`), transitioning *all nine tool schemas* (`get_weather`, `get_time`, `calculator`, `unit_convert`, `database_query`, `fetch_api`, `create_event`, `file_manager`, and `search`) to the entity-agnostic `<THINK>` + 1-hop `<CALL>` pointer architecture achieved **100.0% (26/26) tool routing** and **96.2% (25/26) exact argument extraction**, even routing unseen nonce cities (`"What is the weather in Bloumplik?"` → `get_weather(city="Bloumplik")`). Caveat: Stage 3 was trained on the prepositions and templates used in this 26-case suite, whereas Stage 2 never saw

Table 4: Expanded Random Multi-Carry Extrapolation Sweep from 16 to 100 Digits on CBT-18.95M (`champion_final.pt`, $n = 315$ Problems, Seed 20261022): Each length evaluates a stratified random mixture (70% uniform random digits with multi-carries, 15% dense carries with $a_i + b_i \geq 9$, and 15% random cascade blocks) under deterministic greedy decoding. Because CBT-18.95M was trained on $N_{\text{train}} \leq 15$ digits with shift masking $k = 32$ (Eq. 6), the maximum macro coordinate seen during training is $p_{\max} = 15 + 32 = 47$. CBT-18.95M achieves **100.0% (180/180)** across all tested lengths within the shifted coordinate support ($16 \leq d \leq 40$), **93.3–96.7%** at the $p_{\max} = 47$ boundary ($d = 45, 50$), **90.0%** at $d = 55$, and then breaks down once p exceeds the shifted training support ($d \geq 60$). CPU throughput measured using batched KV-cached FP32 inference on an Intel Core Ultra 5 125H CPU.

Digit Length (d)	Max Column (p)	Coordinate Regime vs. $p_{\max} = 47$	Test Cases (n)	<THINK> Scratchpad Acc	Final Answer Exact Match	95% Wilson CI	(Errors / CPU Tok/s
16 Digits	$p \leq 16$	Within Shifted Support ($p \leq 47$)	30	**100.0% (30/30)**	**100.0% (30/30)**	[88.6, 100.0]%	0 / 328.6 tok/s
20 Digits	$p \leq 20$	Within Shifted Support ($p \leq 47$)	30	**100.0% (30/30)**	**100.0% (30/30)**	[88.6, 100.0]%	0 / 393.5 tok/s
25 Digits	$p \leq 25$	Within Shifted Support ($p \leq 47$)	30	**100.0% (30/30)**	**100.0% (30/30)**	[88.6, 100.0]%	0 / 359.2 tok/s
30 Digits	$p \leq 30$	Within Shifted Support ($p \leq 47$)	30	**100.0% (30/30)**	**100.0% (30/30)**	[88.6, 100.0]%	0 / 265.2 tok/s
35 Digits	$p \leq 35$	Within Shifted Support ($p \leq 47$)	30	**100.0% (30/30)**	**100.0% (30/30)**	[88.6, 100.0]%	0 / 253.6 tok/s
40 Digits	$p \leq 40$	Within Shifted Support ($p \leq 47$)	30	**100.0% (30/30)**	**100.0% (30/30)**	[88.6, 100.0]%	0 / 261.2 tok/s
45 Digits	$p \leq 45$	Near Horizon Boundary ($p \approx 47$)	30	**96.7% (29/30)**	**93.3% (28/30)**	[78.7, 98.2]%	0 / 284.8 tok/s
50 Digits	$p \leq 50$	At Horizon Boundary ($p \approx 47$)	30	**96.7% (29/30)**	**96.7% (29/30)**	[83.3, 99.4]%	0 / 234.7 tok/s
Pooled 16–40d	$p \leq 40$	**Strictly Within** $p_{\max} = 47$	**180**	**100.0% (180/180)**	**100.0% (180/180)**	**[97.9, 100.0]%**	**0 Errors**
Pooled 16–50d	$p \leq 50$	**Full Target Range (16–50d)**	**240**	**99.2% (238/240)**	**98.8% (237/240)**	**[96.4, 99.6]%**	**0 Errors**
55 Digits	$p \leq 55$	Beyond Shifted Support (+8 over $p_{\max}$)	20	**95.0% (19/20)**	**90.0% (18/20)**	[69.9, 97.2]%	0 / 184.2 tok/s
60 Digits	$p \leq 60$	Beyond Shifted Support (+13 over $p_{\max}$)	20	45.0% (9/20)	45.0% (9/20)	[25.8, 65.8]%	3 / 214.4 tok/s
70 Digits	$p \leq 70$	Beyond Shifted Support (+23 over $p_{\max}$)	15	6.7% (1/15)	6.7% (1/15)	[1.2, 29.8]%	0 / 202.7 tok/s
80 Digits	$p \leq 80$	Exceeds $p_{\max}$ & $T = 1024$ ($T \approx 1149$)	10	0.0% (0/10)	0.0% (0/10)	[0.0, 27.8]%	4 / 159.5 tok/s
100 Digits	$p \leq 100$	Exceeds $p_{\max}$ & $T = 1024$ ($T \approx 1429$)	10	0.0% (0/10)	0.0% (0/10)	[0.0, 27.8]%	1 / 141.3 tok/s

Table 5: Stage 3 All-Tools 1-Hop Induction SFT Alignment (`champion_final.pt`) vs. Stage 2 Search-Only SFT (`step_7000`): Extending the entity-agnostic <THINK> + 1-hop <CALL> pointer copy architecture across all 9 tool schemas achieves **100.0% (26/26) tool routing** and **96.2% (25/26) exact argument extraction**, resolving preposition and phrasing sensitivities.

Tool Category	User Prompt (<USER>...)	Stage 2 Search-Only SFT (step_7000)	Stage 3 All-Tools 1-Hop SFT (champion_final.pt)	Status
Weather (on)	What is the weather on Rome	search(query="Roe Weather")	<CALL>get_weather(city="Rome")</CALL>	[PASS]
Weather (in)	What is the weather in Rome?	search(query="the weather")	<CALL>get_weather(city="Rome")</CALL>	[PASS]
Weather (short)	Weather in Tokyo	search(query="Tokyo")	<CALL>get_weather(city="Tokyo")</CALL>	[PASS]
Weather (How is)	How is the weather in Paris	search(query="Ineazhather in Paris")	<CALL>get_weather(city="Paris")</CALL>	[PASS]
Weather (Check)	Check the weather for London	search(query="Looath goston")	<CALL>get_weather(city="London")</CALL>	[PASS]
Weather (Unseen Nonce)	What is the weather in Bloumplik?	search(...)	<CALL>get_weather(city="Bloumplik")</CALL>	[PASS]
Time (in EST)	What is the time in EST	search(query="the time")	<CALL>get_time(timezone="EST")</CALL>	[PASS]
Time (What time)	What time is it in JST	search(query="iis STST")	<CALL>get_time(timezone="JST")</CALL>	[PASS]
Unit (Convert)	Convert 10 km to miles	unit_convert(value=10, ...)	<CALL>unit_convert(value=10, from_unit="km", to_unit="miles")</CALL>	[PASS]
Unit (How many)	How many miles is 25 km	calculator(expr="24*1+...")	<CALL>unit_convert(value=25, from_unit="km", to_unit="miles")</CALL>	[PASS]
Search (Real Entity 1)	What are tomatoes	search(query="tomatoes")	<CALL>search(query="tomatoes")</CALL>	[PASS]
Search (Real Entity 2)	Who is Alexander Graham Bell?	search(query="Alexander Graham Bell")	<CALL>search(query="Alexander Graham Bell")</CALL>	[PASS]
Search (Unseen Nonce 1)	What is a bloumplik	search(query="bloumplik")	<CALL>search(query="bloumplik")</CALL>	[PASS]
Search (Unseen Nonce 2)	Define zaimgromp	search(query="zaimgromp")	<CALL>search(query="zaimgromp")</CALL>	[PASS]
Nested JSON (Database)	Query top 5 active records in engineering.	database_query(...)	<CALL>database_query(filter={"department": "engineering", ...})</CALL>	[PASS]
Nested JSON (API)	Fetch latency metrics for EU from /v2/metrics.	fetch_api(...)	<CALL>fetch_api(endpoint="/v2/metrics/latency", params={...})</CALL>	[PASS]
Direct Conversational	Thank you!	<ANSWER>Hello! How can I...</ANSWER>	<ANSWER>You are very welcome! Let me know if you need anything else.</ANSWER>	[PASS]

them, so the Stage 2 versus Stage 3 comparison is not informative about generalization to new syntax. We did not measure template overlap between the 26 cases and the training generator, and we have no held-out paraphrase set.

3. **Zero-Shot Out-of-Distribution (OOD) Generalization (Table 6)**: On 18 unseen prompts, CBT-18.95M produced well-formed tags for all 18 (18/18) and routed 16/18 (88.9%) to the intended tool. Among the 16 routed cases, **8 (50.0%) copied the argument exactly**; 3 more were near-exact (`"existentialis"`, `"zagnarok"`, `"844 *19"`), giving 11/16 (68.8%) as a secondary, more lenient figure. Because a near-exact string is an incorrect tool argument, exact match is the relevant metric. Exact copies include single words and short lowercase phrases (`"quantum physics"`, `"interstellar"`, `"hyperinflation"`, `"tiramisu"`, `"neutron stars"`, `"Reykjavik"`), but single-word copying is *not* perfect (`"existentialism"`, `"Ragnarok"`, `"theremin"` fail). Multi-word capitalized proper nouns (`"Buenos Aires"` → `"Buris"`, `"Marcus Aurelius"` → `"Arcusnaus"`, `"Kuala Lumpur"` → `"Vuala"`) blend or truncate, plausibly because the synthetic nonce generator trains on single-word pseudowords (a hypothesis we did not test). An embedded digit (`"crispr cas9"`) and a preposition (`"Search for ..."`) each caused a mis-route.

2.2 Isolated Component Ablations of CBT Mechanics (4.24M Token Budget)

Because we tested multiple combinations of CBT components that achieved similar partial extrapolation scores before converging on our final configuration, we conducted a controlled 5-model factor ablation (Table 9) using the exact 100k-parameter budget (`100,672` parameters, $L = 2, d_{\text{model}} = 64, n_{\text{head}} = 4, V = 27$) and the exact same training token volume ($4,236,029$ tokens, 2,000 steps at batch size 32) as the 100k reference model:

1. **1D-RoPE Alone or With Recurrence Fails Completely (Models 1 & 2)**: Under 1D-RoPE, neither $R = 1$ (Model 1, 21.4% overall) nor $R = 3$ re-

Table 6: Zero-Shot Out-of-Distribution (OOD) Generalization of CBT-18.95M (`champion_final.pt`) on 18 Unseen Prompts: 18/18 produced well-formed tags; 16/18 (88.9%) were routed to the intended tool (two drift cases); among the 16 routed cases, **8 (50.0%) copied the argument exactly**, 3 were near-exact (11/16 = 68.8% exact or near-exact), and 5 were partial. Near-exact copies are functionally failures for tool calling: `calculator(expr="844 *19")` computes the wrong product and `search(query="zagnarok")` searches the wrong string. Many prompts reuse phrasing templates seen in training; only the entities are novel.

Unseen OOD Domain	Unseen User Prompt (<USER>...)	Generated Output (<CALL>...</CALL>)	Tool Routing	Character Copy Fidelity
OOD Science	What is quantum physics	<CALL>search(query="quantum physics")</CALL>	[PASS]	[EXACT] (15/15 chars)
OOD Film / Term	What is interstellar	<CALL>search(query="interstellar")</CALL>	[PASS]	[EXACT] (12/12 chars)
OOD Economics	Define hyperinflation	<CALL>search(query="hyperinflation")</CALL>	[PASS]	[EXACT] (14/14 chars)
OOD Cuisine	What is tiramisu?	<CALL>search(query="tiramisu")</CALL>	[PASS]	[EXACT] (8/8 chars)
OOD Astronomy	What are neutron stars	<CALL>search(query="neutron stars")</CALL>	[PASS]	[EXACT] (13/13 chars)
OOD Weather (City)	How is the weather in Reykjavik	<CALL>get_weather(city="Reykjavik")</CALL>	[PASS]	[EXACT] (9/9 chars)
OOD Unit (Inverted)	How many km is 42 miles	<CALL>unit_convert(value=42, from_unit="miles", to_unit="km")</CALL>	[PASS]	[EXACT] (Inverted order parsed)
OOD Timezone	What time is it in GMT	<CALL>get_time(timezone="GMT")</CALL>	[PASS]	[EXACT] (3/3 chars)
OOD Philosophy	What is existentialism?	<CALL>search(query="existentialis")</CALL>	[PASS]	[NEAR-EXACT] (13/14 chars)
OOD Mythology	Tell me about Ragnarok	<CALL>search(query="zagnarok")</CALL>	[PASS]	[NEAR-EXACT] (7/8 chars)
OOD Calculator	Compute 84 * 19	<CALL>calculator(expr="844 *19")</CALL>	[PASS]	[NEAR-EXACT] (6/7 chars)
OOD Deep Learning	What are transformers in deep learning	<CALL>search(query="transwormers of deep")</CALL>	[PASS]	[PARTIAL] (4-word phrase)
OOD Music / Instrument	What is a theremin	<CALL>search(query="ther")</CALL>	[PASS]	[PARTIAL] (prefix copied)
OOD History (2-Word)	Who was Marcus Aurelius?	<CALL>search(query="Arcusnaus")</CALL>	[PASS]	[PARTIAL] (2-word blend)
OOD Weather (2-Word)	What is the weather in Buenos Aires?	<CALL>get_weather(city="Buris")</CALL>	[PASS]	[PARTIAL] (2-word blend)
OOD Weather (2-Word)	Weather on Kuala Lumpur	<CALL>get_weather(city="Vuala")</CALL>	[PASS]	[PARTIAL] (first word copied)
OOD Bio (With Digit)	Explain crispr cas9	<CALL>unit_convert(vanaver="9, ance")</CALL>	[DRIFT]	Digit 9 triggered unit_convert
OOD Architecture	Search for brutalist architecture	<CALL>get_weather(city="Oncordar")</CALL>	[DRIFT]	Preposition for triggered get_weather

Table 7: The Tokenization and Positional Geometry Matrix: Comparing our 2D Matrix Character + Control-Token CBT formulation against 1D Character models, Subword BPE models (4k and 50k vocabularies), Digit-Split 4k BPE, and Learned 2D positional tables, explicitly contrasting **Seen Training Digits** vs. **Unseen Extrapolation Digits**. Vocabulary sizes differ by variant: 115 is the full CBT-18.95M tokenizer, 111 the CBT-200k and Learned-2D variants, and 27 the digit-and-symbol character models. All rows showing '/50' in the Tool Calling column were evaluated on the identical 50-dialogue structured tool suite (6 domains), whereas CBT-18.95M was evaluated on the 26-case 9-schema 1-hop suite of Table 5. Rows were trained with different digit ranges and data and are not a controlled 2×2 study. The unseen-sum column pools 14–20 digits, whereas Table 8 reports 14–16 digits separately (hence 2.3% here and 4.6% there for the 1D character model).

Tokenizer Granularity	Positional Geometry	Vocab (V)	Trained Digits	Tool Calling	Seen Sums (In-Dist)	Unseen Sums (14–20d)	Unseen Cascade Carry
Subword BPE (50k)	1D-RoPE (Flat Line)	50,304	1–12 Digits	N/A (Word-CoT)	99.1% (1–12d)	**0.0%** (Unseen 18–20d)	**0.0%** (Unseen 20d)
Subword BPE (Naive 4k)	1D / 2D-RoPE	4,000	1–12 Digits	**100.0% (50/50)**	**0.0%** (1–12d)	**0.0%** (Unseen 14–20d)	**0.0%** (Unseen 20d)
Digit-Split BPE (4k)	2D-RoPE Matrix	4,114	1–12 Digits	**100.0% (50/50)**	83.3% (1–12d)	64.3% (Unseen 14–20d)	**0.0%** (Unseen 20d)
Atomic Character	1D-RoPE (Flat Line)	27 / 115	1–12 Digits	**100.0% (50/50)**	98.4% (1–12d)	2.3% (Unseen 14–20d)	**0.0%** (Unseen 20d)
Atomic Character	NoPE + Abacus Place	27	1–12 Digits	N/A	**0.0%** (1–12d)	**0.0%** (Unseen 14–20d)	**0.0%** (Unseen 20d)
Atomic Character	Learned 2D Lookup Table	111	1–12 Digits	98.0% (49/50)	100.0% (1–12d)	**0.0%** (Capped $p > 12$)	**0.0%** (Unseen 20d)
CBT-200k (Char + Control)	**2D-RoPE Matrix** ($R = 3$)	**111**	**1–12 Digits**	**100.0% (50/50)**	**100.0% (1–12d)**	**100.0% (Unseen 14–20d)**	**100.0% (Unseen 20d)**
CBT-18.95M (Char + Control)	**2D-RoPE Matrix** ($R = 3$)	**115**	**1–15 Digits**	**100.0% (26/26)**	**100.0% (1–15d)**	**98.8% (237/240 Rand 16–50d)**	**100.0% (Unseen 50d)**

currence (Model 2, 14.3% overall) solves even a single unseen 14–20 digit problem (0.0% extrapolation; loss stalls above `0.13`).

2. **Decoupled 2D-RoPE Is the Primary Factor (Model 1 vs. Model 3)**: The clean single-factor contrast for positional geometry is Model 1 (1D-RoPE, $R = 1, k = 0$) versus Model 3 (Decoupled 2D-RoPE, $R = 1, k = 0$): replacing 1D-RoPE with 2D-RoPE lowers training loss from `0.1864` to `0.0011` and raises accuracy to 100.0% on seen 1–12d and unseen 14–16d sums (78.6%, 11/14 overall).
3. **Recurrence Is Not Needed for Addition in This Ablation (Model 3 vs. Model 4)**: Comparing Model 3 ($R = 1$) against Model 4 ($R = 3$) isolates recurrence: both score 78.6% (11/14) and final loss is slightly worse with $R = 3$ (`0.0017` vs. `0.0011`). We therefore make no claim that recurrent depth contributes to addition at 100k scale. We did not run $R = 1$ versus $R = 3$ at 18.95M, so whether recurrence matters at larger scale or on other tasks is untested.
4. **Shift Masking and Readout Termination (Model 4 vs. Model 5)**: Comparing Model 4 ($k = 0$) and Model 5 ($k = 16$) isolates shift masking on this 14-case set; because they differ on only 1–2 test problems, the comparison is consistent with—but does not statistically establish—a benefit on internal scratchpad columns. At 12.8 epochs (5,000-sample pool), Models 3, 4, and 5 all score 78.6% (11/14) on strict final string match. Inspecting the raw generation tokens of Model 5 (2D-RoPE + $R = 3 + k = 16$) on the 20-digit cascading carry $99\ldots99 + 1$ reveals the following:

<THINK>(0+9+1)=10(1+9+0)=10...(20 cols)...
(1+9+0)=10(1+0+0)=01</THINK>=1$\underbrace{00...00}_{\text{19 zeros}}$<EOS>

Inside <THINK>...</THINK>, Model 5's scratchpad was correct across all columns in the 14 test cases. The error is an *off-by-one readout termination* after </THINK>=, where the model emits 19 zeros instead of 20 before <EOS>. Because the answer-digit coordinates are derived from the scratchpad column count via $p = \max(0, C - 1 - i)$ (Section 3.3), the model itself must decide where the answer ends. Two plausible (untested) hypotheses for this off-by-one error are: (a) the clamp $p = \max(0, C-1-i)$ assigns $p = 0$

Table 8: Cross-Architecture Comparison on Seen vs. Unseen Digit Lengths: Nine 1D/NoPE baselines (the default character model plus eight alternatives, trained on 1–12 digits; several achieve 4.6–18.7% on moderate 14–16 digit extrapolation, while all nine fall to 0.0% on unseen 18–20 digit problems) compared against the 2D-RoPE Character-Based Transformer (CBT) family. CBT-18.95M trained on 1–15 digits and the baselines on 1–12 digits; baselines used no position-offset augmentation and no tuned index hints, so this is not an equalized comparison. The MicroMath/Abacus row (0.0% even in-distribution) likely reflects an under-tuned implementation rather than a property of Abacus embeddings [13].

Architecture Paradigm	Non-Emb Params	Positional Encoding	Trained Digits	Seen In-Dist	Unseen Mod OOD	Unseen Ext OOD	Unseen Extrapolation Summary
Standard Char Transformer	18.91M	1D-RoPE	1–12 Digits	98.4%	4.6% (14–16d)	0.0% (18–20d)	No length beyond 12d reliably
Subword BPE Word-CoT	18.91M	1D-RoPE	1–12 Digits	99.1%	8.0% (14–16d)	0.0% (18–20d)	No length beyond 12d reliably
Philo-Math Co-Trained GPT	18.91M	1D-RoPE	1–12 Digits	94.2%	5.3% (14–16d)	0.0% (18–20d)	No length beyond 12d reliably
Looped Recurrent GPT ($K = 6$)	18.91M	1D-RoPE	1–12 Digits	100.0%	12.0% (14–16d)	0.0% (18–20d)	No length beyond 12d reliably
Hierarchical HRM-1D	18.91M	1D-RoPE	1–12 Digits	100.0%	18.7% (14–16d)	0.0% (18–20d)	No length beyond 12d reliably
Hybrid GLRU-SSM Transformer	18.91M	1D-RoPE + Linear State	1–12 Digits	100.0%	10.7% (14–16d)	0.0% (18–20d)	No length beyond 12d reliably
Harmonic ALU (H-NRM)	18.91M	Poly-Freq Oscillators	1–12 Digits	97.8%	8.0% (14–16d)	0.0% (18–20d)	No length beyond 12d reliably
MicroMath (Abacus NoPE, 1.36M)	1.36M	Discrete Place Only (p)	1–12 Digits	0.0%	0.0% (14–16d)	0.0% (18–20d)	0 Digits (Under-tuned / Collapse)
Foundation HRM-Text (219k tok/s)	1.44M	1D-RoPE + Additive Loop	1–12 Digits	66.7%	0.0% (14–16d)	0.0% (18–20d)	No length beyond 8d reliably
HRM-RoPE (Dual-Loop)	1.56M	**Decoupled 2D-RoPE**	**1–12 Digits**	100.0%	66.7% (14–16d)	**100.0% (18–20d)**	**20 Digits (+8 Unseen Digits)**
CBT-100k ($L = 2, R = 3$)	98.9k	**Decoupled 2D-RoPE**	**1–12 Digits**	**100.0%**	**100.0% (14–16d)**	**100.0% (18–20d)**	**20 Digits (+8 Unseen Digits)**
CBT-200k ($L = 4, R = 3$)	197.8k	**Decoupled 2D-RoPE**	**1–12 Digits**	**100.0%**	**100.0% (14–16d)**	**100.0% (18–20d)**	**20 Digits (+8 Unseen) + Tools**
CBT-18.95M ($L = 6, R = 3$)	**18.89M**	**Decoupled 2D-RoPE**	**1–15 Digits**	**100.0%**	**100.0% (16–40d Rand)**	**95.0% (45–50d); 90% (55d)**	**98.8% (16–50d)** / $p_{\max} = 47$ **Bound**

Table 9: Isolated Component Ablation Study of CBT Mechanics at 100k Scale (100,672 Total Params, Identical 4.24M Token Budget): All 6 models share the exact same 100k character-level backbone ($L = 2, d_{\text{model}} = 64, n_{\text{head}} = 4, V = 27$) and vary three factors indicated in the top header rows: **Positional Encoding** (`1D-RoPE` vs. `Decoupled 2D-RoPE`), **Recurrent Depth** ($R = 1$ single pass vs. $R = 3$ weight-tied passes), and **Coordinate Shift Masking** ($k = 0$ off vs. $k = 16$ random column shift). Models 1–5 are trained for 2,000 steps on a 5,000-sample pool (≈ 12.8 epochs); the 6th column (`CBT-100k Ref`) trains the Model 5 configuration for 64 epochs on a 1,000-sample pool (`dense_100k_2d_rec3_full.pt`). In Model 5, the internal `<THINK>` scratchpad columns were exact in all 14 cases, with the 3 readout misses attributable to an off-by-1 trailing zero before `<EOS>`. Because the 64-epoch reference repeats a 1,000-sample pool 64 times, we cannot rule out memorization or overfitting to boundary patterns on that checkpoint. The 100k ablation uses $k = 16$, whereas CBT-18.95M uses $k = 32$ (Eq. 6). With 14 test cases (where 1–3 cases separate models), Models 3, 4, and 5 cannot be statistically distinguished, and each configuration was trained once.

Benchmark Tier / Configuration	Training Exposure	Model 1 *1D Baseline*	Model 2 *1D + Recurrence*	Model 3 *2D-RoPE Only*	Model 4 *2D + Recurrence*	Model 5 *2D + Rec + Shift*	CBT-100k Ref *2D + Rec + Shift (64 Ep)*
Positional Geometry	—	1D-RoPE (Flat)	1D-RoPE (Flat)	**Decoupled 2D-RoPE**	**Decoupled 2D-RoPE**	**Decoupled 2D-RoPE**	**Decoupled 2D-RoPE**
Recurrent Passes (R)	—	$R = 1$ (2 Layers)	$R = 3$ (6 Eff. Layers)	$R = 1$ (2 Layers)	$R = 3$ (6 Eff. Layers)	$R = 3$ (6 Eff. Layers)	$R = 3$ (6 Eff. Layers)
Random Shift Masking (k)	—	Off ($k = 0$)	Off ($k = 0$)	Off ($k = 0$)	Off ($k = 0$)	**On** ($k = 16$)	**On** ($k = 16$)
Training Pool / Epochs	4.24M Tokens	5k Pool (12.8 Ep)	5k Pool (12.8 Ep)	5k Pool (12.8 Ep)	5k Pool (12.8 Ep)	5k Pool (12.8 Ep)	1k Pool (64.0 Ep)
Final Pretraining Loss	4.24M Tokens	`0.1864`	`0.1353`	`0.0011`	`0.0017`	`0.0017`	`0.0016`
Seen In-Dist (1–12 Digits)	**Seen (1–12d)**	50.0% (3/6)	33.3% (2/6)	**100.0% (6/6)**	**100.0% (6/6)**	**100.0% (6/6)**	**100.0% (6/6)**
Unseen Mod Extrap (14–16d)	**Unseen (+2 to +4d)**	0.0% (0/3)	0.0% (0/3)	**100.0% (3/3)**	**100.0% (3/3)**	**100.0% (3/3)**	**100.0% (3/3)**
Unseen Ext Extrap (18–20d)	**Unseen (+6 to +8d)**	0.0% (0/3)	0.0% (0/3)	33.3% (1/3)	33.3% (1/3)	**66.7% (2/3)**	**100.0% (3/3)**
$10^{17} + 10^{17}$ = (18d Zeros)	Unseen (+6d)	`[FAIL]`	`[FAIL]`	`[FAIL]`	`[FAIL]`	`2e17 [PASS]`	`2e17 [PASS]`
`1234...90+8765...09=` (20d)	Unseen (+8d)	`[FAIL]`	`[FAIL]`	`99...99 [PASS]`	`99...99 [PASS]`	`99...99 [PASS]`	`99...99 [PASS]`
`5555...55+4444...45=` (20d)	Unseen (+8d)	`[FAIL]`	`[FAIL]`	`[FAIL]`	`[FAIL]`	*100% Think / Off-by-1 0*	`1e20 [PASS]`
Unseen 20d Zeros & Carry	**Unseen (+8d)**	0.0% (0/2)	0.0% (0/2)	50.0% (1/2)	50.0% (1/2)	*100% Think / Off-by-1 0*	**100.0% (2/2)**
$10^{19} + 10^{19}$ = (20d Zeros)	Unseen (+8d)	`[FAIL]`	`[FAIL]`	`2e19 [PASS]`	`2e19 [PASS]`	*19/19 Cols OK, 18 0s out*	`2e19 [PASS]`
$99\ldots99 + 1$ = (20d Carry)	Unseen (+8d)	`[FAIL]`	`[FAIL]`	`[FAIL]`	`[FAIL]`	*21/21 Cols OK, 19 0s out*	`1e20 [PASS]`
Internal `<THINK>` Math Acc	All 14 Cases	21.4% (3/14)	14.3% (2/14)	78.6% (11/14)	78.6% (11/14)	**100.0% (14/14 Exact)**	**100.0% (14/14)**
Strict Final String Match	All 14 Cases	21.4% (3/14)	14.3% (2/14)	78.6% (11/14)	78.6% (11/14)	78.6% (11/14)	**100.0% (14/14)**

to both the true units digit ($i = C - 1$) and any extra overflow digit ($i \geq C$), making the last answer digit and the post-answer position share the same macro coordinate $p = 0$; and (b) after shift masking moves $p > 0$ by $\Delta \in [0, 16]$ while $p = 0$ stays fixed, the $p = 1 \to p = 0$ boundary is under-trained at 12.8 epochs. When the exact same architecture is trained for 64 epochs on a 1,000-sample pool (`CBT-100k Ref`, 4.24M tokens) or scaled to **CBT-18.95M** (247.3M tokens), the readout error disappeared in our tests (14/14 for `CBT-100k Ref`; 20/20 at 16–50 digits for CBT-18.95M), though repeating a 1,000-sample pool 64 times raises the possibility of overfitting to training length distributions rather than purely resolving boundary attention.

2.3 Strictly Equalized 1–15 Digit, $T = 1024$ Comparison with Ruoss et al. Randomized Positions (12,800 Random Problems)

Because the 18.91M 1D baselines in Table 8 were trained on 1–12 digits with a 512-token context window and without position-offset augmentation, we conducted a strictly equalized experiment (`experiments/run_equalized_15d_1024_and_large_sweep.py`, Table 10) where all models share the exact same 100k character-level architecture ($L = 2, d_{\text{model}} = 64, n_{\text{head}} = 4, R = 1, V = 27$), the exact same 6,000-sample **1–15 digit** training set, a **1024-token context window**, and 3,000 training steps at batch size 32, and evaluated each model on **200 stratified random multi-carry problems at every length** $(8, 12, 15, 16, 20, 25, 30, 35, 40, 45, 50, 55, 60, 70, 80, 100$

Table 10: Strictly Equalized 1–15 Digit, $T = 1024$ Controlled Comparison Including Ruoss et al. (2023) Randomized Position Encodings Evaluated on 200 Random Multi-Carry Problems per Length (3,200 Problems/Model, 12,800 Total): All four models share the exact same 100k character-level backbone ($L = 2, d_{\text{model}} = 64, n_{\text{head}} = 4, R = 1, V = 27$), the exact same 6,000-equation **1–15 digit** training dataset (matching CBT-18.95M's 1–15d distribution), a **1024-token context window**, and 3,000 training steps at batch size 32. Neither standard 1D-RoPE, 1D-RoPE with random translation offset ($\Delta \sim \mathcal{U}\{0..512\}$), nor 1D-RoPE with Ruoss et al. [16] randomized positions ($0 \le t_1 < \cdots < t_T < 1024$) solves any unseen 16–100 digit addition (0/200, 95% Wilson CI [0.0, 1.9]%), whereas equalized Decoupled 2D-RoPE ($k = 32, p_{\max} = 47$) reaches **92.0% (184/200)** at 16d and 25d, **90.0% (180/200)** at 20d, and **85.0% (170/200)** at 30d before dropping to 0.0% beyond $p_{\max} = 47$ ($d \ge 50$).

Evaluation Digit Length ($n = 200$/**len**)	**Exposure Status**	**1D-RoPE Standard** *1–15d, $T = 1024$*	**1D-RoPE + Offset** $\Delta \sim \mathcal{U}\{0..512\}$	**1D-RoPE + Ruoss et al. [16]** *Rand Pos $M_{\max} = 1024$*	**Decoupled 2D-RoPE + Shift** *Ours (100k, $R = 1, k = 32$)*	**CBT-18.95M Ref** ($L = 6, R = 3$) *Ours (18.95M, $R = 3, k = 32$)*
Final Training Loss (3,000 Steps)	1–15 Digits	0.09822	0.09582	0.23179	0.00597	0.00009
Seen 8, 12, 15 Digits ($n = 600$)	Seen (1–15d)	0.0% (0/600) [0.0, 0.6]%	0.0% (0/600) [0.0, 0.6]%	0.0% (0/600) [0.0, 0.6]%	**69.5% (417/600)** [65.7, 73.1]%	**100.0%** (Seen 1–15d)
of which 15 Digits ($n = 200$)	Max Seen (15d)	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	**86.0% (172/200)** [80.5, 90.1]%	**100.0%**
Unseen 16 Digits ($n = 200$)	**Unseen (+1d)**	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	**92.0% (184/200)** [87.4, 95.0]%	**100.0% (30/30)** [88.6, 100]%
Unseen 20 Digits ($n = 200$)	**Unseen (+5d)**	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	**90.0% (180/200)** [85.1, 93.4]%	**100.0% (30/30)** [88.6, 100]%
Unseen 25 Digits ($n = 200$)	**Unseen (+10d)**	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	**92.0% (184/200)** [87.4, 95.0]%	**100.0% (30/30)** [88.6, 100]%
Unseen 30 Digits ($n = 200$)	**Unseen (+15d)**	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	**85.0% (170/200)** [79.4, 89.3]%	**100.0% (30/30)** [88.6, 100]%
Unseen 35 Digits ($n = 200$)	**Unseen (+20d)**	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	**78.0% (156/200)** [71.8, 83.2]%	**100.0% (30/30)** [88.6, 100]%
Unseen 40 Digits ($n = 200$)	**Unseen (+25d)**	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	**51.5% (103/200)** [44.6, 58.3]%	**100.0% (30/30)** [88.6, 100]%
Unseen 45 Digits ($n = 200$)	**Unseen (+30d)**	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	0.0% (0/200) [0.0, 1.9]%	**30.0% (60/200)** [24.1, 36.7]%	**93.3% (28/30)** [78.7, 98.2]%
Unseen 50–100 Digits ($n = 1{,}200$)	**Beyond** $p_{\max} = 47$	0.0% (0/1200) [0.0, 0.3]%	0.0% (0/1200) [0.0, 0.3]%	0.0% (0/1200) [0.0, 0.3]%	0.0% (0/1200) [0.0, 0.3]%	96.7% (50d), 90% (55d), 45% (60d)
Pooled Unseen 16–40d ($n = 1{,}200$)	**Within** $p_{\max} = 47$	0.0% (0/1200) [0.0, 0.3]%	0.0% (0/1200) [0.0, 0.3]%	0.0% (0/1200) [0.0, 0.3]%	**81.4% (977/1200)** [79.1, 83.5]%	**100.0% (180/180)** [97.9, 100]%

digits; 3,200 problems per model, 12,800 problems total):

1. **Equalized 1D-RoPE and 1D Random Offset Fail Beyond Short Sums**: Even when trained on 1–15 digits with a 1024-token context window, standard 1D-RoPE (`loss=0.0982`) and 1D-RoPE with random translation offsets $\Delta \sim \mathcal{U}\{0, \ldots, 512\}$ (`loss=0.0958`) solve 2-digit sums (`15+25=40`) but score **0.0% (0/200; 95% Wilson CI** $[0.0, 1.9]\%$) at every random multi-carry length from 8 to 100 digits.
2. **Why Ruoss et al. [16] Randomized 1D Positions Do Not Substitute for 2D Column Alignment**: Sampling sorted random 1D positions $0 \le t_1 < t_2 < \cdots < t_T < 1024$ during training causes training loss to stall at `0.2318` and yields **0.0% (0/200)** across all tested lengths. In a flat 1D sequence without explicit index hints, the model relies on consistent relative token distances to locate corresponding operand digits; randomizing inter-token 1D gaps destroys those relative offsets, whereas shifting the macro coordinate $p \leftarrow p + \Delta$ in Decoupled 2D-RoPE preserves $\Delta p = p_t - p_j = 0$ between every scratchpad column and its matching operand digits.
3. **Equalized Decoupled 2D-RoPE ($k = 32$) Extrapolates Across 1,200 Unseen 16–40d Problems**: Under the exact same 100k $R = 1$ architecture and 3,000-step budget, Decoupled 2D-RoPE drops training loss to `0.0060` and achieves **81.4% (977/1,200; 95% Wilson CI** $[79.1, 83.5]\%$) pooled across unseen 16–40 digit random multi-carry problems—including **92.0% (184/200)** at 16d and 25d, **90.0% (180/200)** at 20d, and **85.0% (170/200)** at 30d—before dropping at 45d (**30.0%, 60/200**) and reaching 0.0% at $d \ge 50$ once column indices exceed the $p_{\max} = 15 + 32 = 47$ shift-masking support.

Table 11: Exact Layer-by-Layer Parameter Topology of CBT-18.95M: $d_{\text{model}} = 512$, $n_{\text{head}} = 8$ ($d_k = 64$), $L = 6$ physical blocks, $R = 3$ recurrent passes ($L_{\text{eff}} = 18$), bias disabled, tied input/output embeddings.

Component / Module	**Tensor Shape**	**Parameters**	**% of Total**
Token Embedding (W_{te})	115×512	58,880	0.31%
Per Physical Block ($l \in \{1..6\}$)**:**			
Pre-Attn LayerNorm (`ln_1`)	2×512	1,024	0.005%
Fused QKV Proj (`c_attn`)	512×1536	786,432	4.15%
Attention Output Proj (`c_proj`)	512×512	262,144	1.38%
Pre-MLP LayerNorm (`ln_2`)	2×512	1,024	0.005%
MLP Up-Proj (`c_fc`, 4×)	512×2048	1,048,576	5.54%
MLP Down-Proj (`mlp.c_proj`)	2048×512	1,048,576	5.54%
Subtotal per Physical Block	—	*3,147,776*	*16.62%*
All 6 Physical Blocks ($L = 6$)	$6 \times 3{,}147{,}776$	18,886,656	99.69%
Recurrent Unrolling ($R = 3$)	Weight-Tied ($L_{\text{eff}} = 18$)	+0	0.00%
Decoupled 2D-RoPE Cache	Non-Parametric (cos, sin)	+0	0.00%
Final LayerNorm (`ln_f`)	2×512	1,024	0.005%
LM Output Head ($W_{\text{lm_head}}$)	Tied to W_{te}	+0	0.00%
Non-Embedding Parameters	—	**18,887,680**	**99.69%**
Total Model Parameters	—	**18,946,560**	**100.00%**

3 Architecture, 2D Matrix Embeddings, Tokenizer, and Training Data

In this section, we provide a complete technical specification of **CBT-18.95M**, detailing its hybrid character-control tokenizer, Decoupled 2D-RoPE embeddings, weight-tied recurrent depth, and the exact composition of the training datasets.

3.1 Atomic Character and Structural Control Tokenizer ($V = 115$)

To eliminate BPE subword chunking distortions while retaining clear structural boundaries for multi-turn agentic dialogues, our tokenizer (`ScaledTokenizer`) defines a compact vocabulary $\mathcal{V}$ of size $V = 115$ for CBT-18.95M (the smaller CBT-200k and 100k variants use character subsets with $V = 111$ and $V = 27$ respectively), parti-

tioned into two disjoint subsets:

$$\mathcal{V} = \mathcal{V}_{\text{ctrl}} \cup \mathcal{V}_{\text{char}}, \quad |\mathcal{V}_{\text{ctrl}}| = 14, \quad |\mathcal{V}_{\text{char}}| = 101 \tag{3}$$

1. **Structural Control Tokens ($\mathcal{V}_{\text{ctrl}}$, IDs $0 \dots 13$)**: Consists of 4 standard sequence control tokens (`<PAD>`, `<BOS>`, `<EOS>`, `<UNK>`) and 10 paired discourse state delimiters that structure agentic reasoning:

   `<USER>`, `</USER>`, `<THINK>`, `</THINK>`, `<CALL>`, `</CALL>`, `<RESULT>`, `</RESULT>`, `<ANSWER>`, `</ANSWER>`

2. **Atomic ASCII Characters ($\mathcal{V}_{\text{char}}$, IDs $14 \dots 114$)**: Contains all printable ASCII characters—lowercase letters (`a--z`), uppercase letters (`A--Z`), single digits (`0--9`), punctuation symbols (`+`, `=`, `(`, `)`, `{`, `}`, `"`, etc.), space, newline, and tab.

During tokenization, a greedy left-to-right prefix scanner checks if the input substring at index i matches any tag in $\mathcal{V}_{\text{ctrl}}$. If matched, the entire tag (e.g., `<THINK>`) is emitted as a single token ID; otherwise, the single character $s[i]$ is emitted as its atomic character ID.

3.2 Decoupled 2D Rotary Position Embeddings (2D-RoPE)

In **Decoupled 2D-RoPE**, every token x_t is associated with a 2D coordinate pair $(p_t, \tau_t) \in \mathbb{N}^2$, where p_t is the **Macro (Row) Coordinate** and τ_t is the **Micro (Column) Coordinate**. Within each attention head of dimension $d_k = 64$, we split the query and key vectors into two orthogonal 32-dimensional subspaces:

$$\tilde{\mathbf{q}}_t^{(h)} = \begin{bmatrix} \mathcal{R}_\Theta(p_t)\, \mathbf{q}_{t,:32}^{(h)} \\ \mathcal{R}_\Theta(\tau_t)\, \mathbf{q}_{t,32:}^{(h)} \end{bmatrix}, \quad \tilde{\mathbf{k}}_t^{(h)} = \begin{bmatrix} \mathcal{R}_\Theta(p_t)\, \mathbf{k}_{t,:32}^{(h)} \\ \mathcal{R}_\Theta(\tau_t)\, \mathbf{k}_{t,32:}^{(h)} \end{bmatrix} \tag{4}$$

For any coordinate scalar $c \in \{p_t, \tau_t\}$ and a 32-dimensional subspace vector $\mathbf{u} \in \mathbb{R}^{32}$, $\mathcal{R}_\Theta(c)\mathbf{u}$ applies 2×2 planar rotations across the 16 channel pairs $(2j, 2j+1)$ for $j \in \{0, 1, \dots, 15\}$ with $\theta_j = 10000^{-2j/32}$.

Proposition 1 (Zero Phase Drift & Scale-Invariant Column Alignment)**.** *Let q_t be a scratchpad query token in decimal column p and k_j be the corresponding operand digit in column p. Under Decoupled 2D-RoPE (Eq. 4), the pre-softmax attention logit $\langle \tilde{\mathbf{q}}_t^{(h)}, \tilde{\mathbf{k}}_j^{(h)} \rangle / \sqrt{d_k}$ is strictly independent of the total number of digits N:*

$$\begin{aligned} \left(\tilde{\mathbf{q}}_t^{(h)}\right)^\top \tilde{\mathbf{k}}_j^{(h)} = \left(\mathbf{q}_{t,:32}^{(h)}\right)^\top \mathbf{k}_{j,:32}^{(h)} \\ + \left(\mathbf{q}_{t,32:}^{(h)}\right)^\top \mathcal{R}_\Theta(\tau_t)\, \mathbf{k}_{j,32:}^{(h)} \end{aligned} \tag{5}$$

Proof. Because both the scratchpad token and the prompt operand digit belong to decimal column 10^p, $p_t = p_j = p \implies \Delta p = 0 \implies \mathcal{R}_\Theta(0) = \mathbf{I}_{32}$, while $\tau_t \in \{0, \dots, 9\}$ is strictly bounded for all $N \in \mathbb{N}$. □

The proposition is conditional on the coordinates being assigned correctly: it states that a single logit between a scratchpad query token and its corresponding operand digit does not depend on N, not that the model computes carries correctly or that the coordinates are produced by the model. Moreover, Proposition 1 covers only scratchpad-to-operand attention; for the final answer digits after `</THINK>=`, the parser sets $p = \max(0, C - 1 - i)$ using the model's emitted scratchpad column count C, so answer-to-scratchpad alignment depends on the model having emitted the right number of scratchpad columns. Since RoPE depends only on relative offsets, the reason shift masking is considered during training is not absolute position but the large offset between scratchpad rows ($p > 0$) and the prompt tokens at $p = 0$ that appears at test length N but was never seen at training length.

3.3 Coordinate Assignment at Inference

The coordinates are computed by a deterministic, hand-written rule-based parser (`compute_2d_coords` in the `scaled_t4` model file) that is called inside the forward pass over the token ids; it is applied identically in training and in autoregressive generation, and the model does not predict (p, τ). The rules are:

- *Operands*: a digit in the first (second) operand gets p equal to its distance from the `+` (`=`) token minus one, so units digits have $p = 0$; $\tau = 0$.
- *Scratchpad*: inside `<THINK>`, each `(` token starts a new column (p increments, τ resets to 0), and τ counts tokens since the `(`, capped at 9. The model therefore decides when to advance a column by emitting `(`; the parser does not count 10-character templates. Because column advancement is keyed on `(`, the model can advance p only by emitting `(`, which also means a single spurious or missing `(` would shift all subsequent column coordinates. Across all 240 random 16–50 digit problems (Table 4), the 20 structured 16–50 digit problems, and the 14 ablation cases (Models 3–5 and `CBT-100k Ref`), **zero errors arose from a misemitted `(` inside `<THINK>`**; `(` column-count errors appeared only at $d \geq 60$ once macro coordinates exceeded the shift-masked training horizon $p_{\max} = 47$ (3/20 cases at 60d, 4/10 at 80d, 1/10 at 100d). The 1D baselines, by contrast, failed by repeating valid 10-character `(0+0+0)=00` blocks too many times rather than corrupting the `(` delimiter syntax.
- *Answer*: after `</THINK>=`, the i-th digit emitted gets $p = \max(0, C-1-i)$ where C is the number of scratchpad columns. The answer length is not given to the model; extra digits are clamped at $p = 0$, so the unit digit ($i = C - 1$) and any extra overflow digit ($i \geq C$) receive the same coordinate $p = 0$. The model must therefore identify the termination point itself, and this $p = 0$ clamp—alongside the $p = 1 \to p = 0$ shift-

masking boundary—is a plausible cause of the off-by-one readout error in Table 9.

- *Discourse mode*: if `<USER>` is present, $p \in \{0, \ldots, 4\}$ is the index of the most recent segment tag (`<USER>`, `<THINK>`, `<CALL>`, `<RESULT>`, `<ANSWER>`) and τ counts tokens since that tag.

Thus the operand-to-column alignment (the operand and column ranks) is supplied by the parser, and the length-generalization result should be read as conditional on this hand-designed alignment. The parser also assumes a fixed prompt format (`A+B=`), and is a Python loop over the batch that is not optimized for speed.

During training, we apply **Selective Coordinate Shift Masking**:

$$p_t \leftarrow p_t + \Delta \cdot \mathbb{I}(p_t > 0), \quad \Delta \sim \mathcal{U}\{0, 1, \ldots, k\} \tag{6}$$

with $k = 32$ for CBT-18.95M and $k = 16$ for the 100k ablation models of Table 9. This exposes $p > 0$ arithmetic columns to indices up to $p = 47$ during 15-digit training (with $k = 32$), while keeping the `<USER>` prompt ($p = 0$) anchored at the origin. It is closely related to randomized positional encodings [16].

3.4 Weight-Tied Recurrent Depth with $1/\sqrt{R}$ Scaling

We unroll the $L = 6$ physical Transformer blocks across $R = 3$ recurrent passes per forward step ($L_{\text{eff}} = 18$ layers at zero parameter overhead) using a $1/\sqrt{R}$-scaled residual update:

$$\mathbf{x} \leftarrow \mathbf{x} + \frac{1}{\sqrt{R}} \left(\text{Block}_l(\mathbf{x}, p, \tau) - \mathbf{x}\right) \tag{7}$$

The $1/\sqrt{R}$ factor is a heuristic residual damping choice. We do not derive a variance bound for it, and our ablation (Table 9) shows no benefit from recurrence for addition. We therefore drop the variance comparison between $1/\sqrt{R}$ scaling and unscaled accumulation that appeared in earlier drafts.

3.5 Three-Stage Training Data & 1-Hop Induction Curriculum

Table 12 summarizes the three-stage training protocol that produced `champion_final.pt` on dual NVIDIA T4 GPUs (FP16 AMP):

1. **Stage 1 (Joint Pretraining, 45,059 steps, 247.3M tokens, Loss `0.000094`)**: Trained on a 50%/50% mixture of 1–15 digit column addition (40% asymmetric lengths, 25% symmetric lengths, 20% cascading carries $99\ldots9 + b$, 15% power-of-ten zero chains) and 6-category structured tool dialogues, with -100 loss masking on prompts and `<RESULT>` payloads.
2. **Stage 2 (Search Copy Induction SFT, 7,000 steps, Loss `0.004834`)**: In early SFT tests, repeating an unfamiliar entity X inside `<THINK>` before calling `<CALL>` caused a *2-Hop Copy Bottleneck*, where character n-gram priors hallucinated phonotactic nonces (`"tomatoes"` → `"smoom"`, `"bloumplik"` → `"cheat"`). Decoupling `<THINK>` into an invariant intent statement and training on 65% procedurally generated phonotactic pseudowords ($L_{\text{memorize}} \to \infty$) forced a **1-hop induction pointer head** ($p = 2 \to p = 0$) to copy unseen strings character-by-character.
3. **Stage 3 (All-Tools 1-Hop Annealing SFT, 4,000 steps, LR $5 \times 10^{-5} \to 5 \times 10^{-6}$, Loss `0.007532`)**: Extended the 1-hop pointer copy formulation across all 9 tool schemas (`get_weather`, `get_time`, `calculator`, `unit_convert`, `database_query`, `fetch_api`, `create_event`, `file_manager`, and `search`) alongside a 10% 1–15 digit math replay buffer, achieving **100.0% (26/26) tool routing**, **96.2% (25/26) exact argument extraction**, and **100.0% 1–50 digit sum extrapolation**.

3.6 Hyperparameters and Release

Defaults in the training code (`train_and_eval.py` and `sft_search.py` under `champion_model/scaled_t4/`): AdamW with weight decay 0.01, peak learning rate 10^{-3} with cosine decay, gradient-norm clipping at 1.0, per-GPU batch size 32 on 2 GPUs, context length (block size) 1024 tokens with up to 4096 coordinate indices, $k = 32$ shift range, $R = 3$, FP16 autocast with gradient scaling. Stage 3 SFT uses a learning rate from 5×10^{-5} to 5×10^{-6}. Context length is not the limiting factor at 50 digits (about 10 characters per scratchpad column gives roughly 600 tokens), but the 1D baselines were trained with a 512-token block, so their 20+ digit failures are partly confounded with context length; their failures at 14–16 digits are not. Arithmetic and tool data are produced by procedural generators in the repository (`src/arithmetic.py` and the dataset modules); the code, data generators, and checkpoints are available in the project repository.

4 Discussion, Limitations, and Future Horizons

4.1 Positioning Within Prior Length-Generalization Work

Prior studies have shown that standard Transformers fail to extrapolate arithmetic beyond training lengths under 1D positional encodings [5, 7, 8, 14]. Several lines of work address this by modifying positional information or data formatting: Shen et al. [6], Zhou et al. [17], and Duan &

Table 12: Complete Three-Stage Training Data Composition, Digit Horizon Boundaries, and Loss Masking Specification for CBT-18.95M (`champion_final.pt`): Across all stages, arithmetic training is strictly bounded to **1–15 digits** ($N \leq 15$). All 16- to 50-digit evaluations are 100% zero-shot out-of-distribution extrapolations.

Training Stage	Dataset Component / Sub-Distribution	Pattern / Schema Specification	Trained Digit / Length Scope	Mixture Weight	Supervised Target (Masking Rule)
Stage 1: Pretraining	**A. Arithmetic State-Machine Corpus (50%)**	$A + B =$ <THINK>$(c + a + b) = c's \ldots$</THINK>$= S$	**Strictly 1–15 Digits** ($N \leq 15$)	**50.0% of Pretrain**	Prompt $A + B$ = masked (-100)
	1. Asymmetric & Symmetric Random Sums	$d_A, d_B \sim \mathcal{U}[1, 15]$ (40% asymmetric, 25% symmetric)	$1 \leq d_A, d_B \leq 15$ (0% $>$ 15d)	32.5% (65% of Math)	Scratchpad + Final Sum supervised
	2. Cascading Carry & Zero Chains	$99 \ldots 9 + b$ (20%) and $k_1 10^{d_1 - 1} + k_2 10^{d_2 - 1}$ (15%)	$2 \leq d \leq 15$ (0% $>$ 15d)	17.5% (35% of Math)	Scratchpad + Final Sum supervised
	B. Multi-Domain Tool Corpus (50%)	<USER> → <THINK> → <CALL> → <RESULT> → <ANSWER>	$T \leq 512$ Characters	**50.0% of Pretrain**	<USER> & <RESULT> masked (-100)
	Single-Tool, Chaining, JSON, Error, Chat	6 interaction categories across cities, timezones, units	1–3 Conversational Turns	50.0% (6 Equal Cats)	<THINK>, <CALL>, <ANSWER>
Stage 2: Search SFT	**Open-Domain Search 1-Hop Copy (7k steps)**	Entity-agnostic <THINK> + `search(query="X")`	65% Dynamic Nonces / 35% Real	**81.25% (26/32 Batch)**	Forces $L_{\text{memorize}} \to \infty$ pointer copy
	Math & Multi-Tool Replay	1–15d carry chains + tool schemas	**Strictly 1–15 Digits** ($N \leq 15$)	**18.75% (6/32 Batch)**	Preserves 50d math & tool syntax
Stage 3: All-Tools SFT	**Universal 9-Tool 1-Hop Annealing (4k steps)**	All 9 tools (`weather, time, unit, calc, db, api, search...`)	Varied Prepositions & Nonces	**90.0% of Stage 3**	Resolves preposition/routing drift
	High-Digit Arithmetic Carry Replay	Cascading carries & random sums ($N \leq 15$d)	**Strictly 1–15 Digits** ($N \leq 15$)	**10.0% of Stage 3**	Maintains 100.0% 1–50d math

Shi [18] use explicit index hints and structured scratchpads; Ruoss et al. [16] randomize positional indices across a wider range during training (closely related to our shift masking in Eq. 6); McLeish et al. [13] introduce Abacus embeddings that align digits of equal place value; and Fan et al. [19] use looped Transformers for iterative algorithms. Our Decoupled 2D-RoPE is philosophically similar to Abacus embeddings and explicit index hints in that it supplies place-value alignment (p, τ) via a rule-based parser rather than learning it from raw text, while extending the coordinate grid to a two-dimensional (p, τ) RoPE subspace that accommodates both multi-character column scratchpads and multi-segment tool dialogues.

4.2 Why CBT Suits Local Edge Inference Rather Than Brute-Force Scaling

A central takeaway of our work is that **CBT is not meant to replace subword tokenization in massive trillion-parameter cloud LLMs as-is**. Because CBT generates text *character-by-character* and executes $R = 3$ recurrent passes per character, generating a 50-word paragraph requires ~ 300 sequential forward passes (compared to ~ 65 subword passes in a BPE model). At billion-parameter scale, this character-by-character serial bottleneck would make inference prohibitively slow.

However, at compact scales (**100k to 18.95M parameters**), individual forward passes involve a small matrix footprint (36 MB of FP16 weights, though $R = 3$ recurrence triples the per-token FLOPs to roughly that of a $\sim$57M-parameter dense model and character-level sequences expand the KV cache). On an Intel Core Ultra 5 125H laptop CPU (FP32), batched KV-cached decoding runs at **141–394 character tokens/sec** across 16–100 digit traces (and 1.1–5.3 tokens/sec uncached single-stream); we did not benchmark mobile ARM hardware or INT8 quantization, nor did we compare against fine-tuned small off-the-shelf subword models on tool calling. Subject to those practical constraints, CBT's ability to handle intent classification, 9-schema tool routing (26/26 on template-matched prompts; 8/16 exact argument copying on OOD prompts), and 16–50 digit sums (237/240 random; 20/20 structured) within a 36 MB weight file suggests potential utility for:

- **Low-Resource Local Tool Routers**: On-device assistants on mobile phones, embedded IoT controllers, or local desktop daemons that parse user commands into structured API/JSON calls (`get_weather`, `file_manager`, `database_query`) without cloud connectivity.
- **Deterministic Symbolic Co-Processors**: Compact local modules that execute exact character-level formatting, arithmetic, and pointer extraction without subword hallucination.

4.3 Open Questions and Further Testing

While CBT-18.95M shows that placing characters on a hand-assigned 2D coordinate grid (p, τ) yields 100.0% accuracy (180/180) on random 16–40 digit sums, 98.8% (237/240) across 16–50 digits, and 90.0% (18/20) at 55 digits before degrading beyond the $p_{\max} = N_{\text{train}} + k = 47$ shift-masking horizon (Table 4), our OOD and ablation experiments leave several questions open:

1. **Sensitivity to Epoch Reinforcement and Boundary Coordinates**: As shown in our 100k component ablations (Table 9), while several combinations of 2D-RoPE and recurrence achieved identical 78.6% strict match at 12.8 epochs, the off-by-one trailing zero readout error at the $p = 1 \to p = 0$ boundary disappeared in our runs with higher epoch density (64 epochs on a 1,000-sample pool) or larger capacity (18.95M); however, these single-run comparisons do not isolate whether the fix came from boundary exposure, memorization of the small pool, or model scale.
2. **Multi-Word OOD Pointer Spans and General Algorithms**: In zero-shot OOD testing (Table 6), two-word capitalized proper nouns (`"Buenos Aires"` → `"Buris"`) still experience partial compression, and our algorithmic generalization scope here focused on $\mathcal{O}(N)$ column sums rather than $\mathcal{O}(N^2)$ multiplication or arbitrary graph algorithms.
3. **Learned Coordinates and Other Tasks**: Emitting hundreds of scratchpad characters is an inefficient simulation of an ALU, the coordinates are hand-assigned, and extrapolation is bounded by the shift-masked coordinate support $p_{\max} = N_{\text{train}} + k$ (Table 4). Important next steps are learning or infer-

ring (p, τ), testing on a second task whose coordinates are not trivial (multiplication, sorting, string reversal), running multi-seed training, and comparing against equalized 1D baselines with randomized positions, index hints, and tuned Abacus embeddings [16, 17, 18, 19].

5 Conclusion

Starting from the failure of a standard 18.91M-parameter Transformer and eight architectural baselines on unseen 18- to 20-digit sums (**0.0% extrapolation at 18–20 digits**, despite 4.6–18.7% on moderate 14–16 digit sums), our experiments suggest that position geometry is the dominant factor (character-level 1D-RoPE: 2.3% across 14–20d; character-level 2D: 100%), with subword tokenization secondary. We studied the **Character-Based Transformer (CBT)**, which generates atomic characters and structural control tags on a two-dimensional coordinate matrix (p, τ) via **Decoupled 2D-RoPE** and weight-tied recurrent depth ($R = 3$, which our ablation found unnecessary for addition). Trained strictly on 1- to 15-digit sums with $k = 32$ shift masking ($p_{\max} = 47$) and multi-tool dialogues, **CBT-18.95M** (`champion_final.pt`) solves 20/20 structured 16–50 digit sums, achieves **180/180 (100.0%)** on random multi-carry sums across 16–40 digits and **237/240 (98.8%)** across 16–50 digits (degrading beyond $p_{\max} = 47$ from 90.0% at 55d to 45.0% at 60d and 0.0% at 80–100d), routes 26/26 tool cases (25/26 exact arguments, on templates shared with training), and routes 16/18 OOD prompts, of which 8/16 (50.0%) have exact arguments. Recurrent depth gave no measurable benefit for addition in our 100k ablation. Ultimately, our findings show that while 2D embeddings and character tokenization achieve high performance with exceptional **memory efficiency**, character-by-character recurrence is **not compute-efficient** at scale—meaning current subword LLMs remain well-suited for general cloud AI, whereas CBT is a candidate design for memory-constrained local tool-calling agents and a starting point for studying coordinate-aware architectures, pending the broader tests listed above.

Disclaimer: Use of AI and Compute Infrastructure

Use of AI Assistance: Portions of the experimental engineering, code implementation, data pipeline structuring, and manuscript drafting/formatting in this research were conducted with the assistance of **Gemini 3.8 Flash**. All experimental hypotheses, architectural selections, empirical benchmarks, and final scientific conclusions were verified and validated by the author.

Compute Hardware: All model pretraining, supervised fine-tuning (SFT), and isolated component ablation runs reported in this paper were executed on **Kaggle's NVIDIA Tesla T4 GPUs** (16 GB GDDR6 VRAM, Turing `sm_75` architecture) using PyTorch native FP16 automatic mixed precision (`torch.float16`).

References

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

[2] J. Su, M. Ahmed, Y. Lu, S. Pan, W. Bo, and Y. Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.

[3] J. Hoffmann, S. Borgeaud, A. Mensch, E. Buchatskaya, T. Cai, E. Rutherford, D. de Las Casas, L. A. Hendricks, J. Welbl, A. Clark, et al. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 2022.

[4] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 1877–1901, 2020.

[5] R. Nogueira, Z. Jiang, and J. Lin. Investigating the limitations of transformers with simple arithmetic tasks. *arXiv preprint arXiv:2102.13019*, 2021.

[6] R. Shen, S. Bubeck, R. Eldan, Y. T. Lee, Y. Li, and Y. Zhang. Positional description matters for transformers arithmetic. *arXiv preprint arXiv:2311.14737*, 2023.

[7] S. Jelassi, S. d'Ascoli, C. Domingo-Enrich, Y. Wu, Y. Li, and F. Charton. Length generalization in arithmetic transformers. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

[8] N. Dziri, X. Lu, M. Sclar, X. L. Li, L. Jiang, B. Y. Lin, S. Welleck, P. West, C. Bhagavatula, R. Le Bras, et al. Faith and fate: Limits of transformers on compositionality. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

[9] M. Nye, A. J. Andreassen, G. Gur-Ari, H. Michalewski, J. Austin, D. Bieber, D. Dohan, A. Lewkowycz, M. Bosma, D. Luan, et al. Show your work: Scratchpads for intermediate computation with language models. *arXiv preprint arXiv:2112.00114*, 2021.

[10] J. Wei, X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q. V. Le, D. Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 24824–24837, 2022.

[11] M. Dehghani, S. Gouws, O. Vinyals, J. Uszkoreit, and Ł. Kaiser. Universal transformers. In *International Conference on Learning Representations (ICLR)*, 2019.

[12] A. Giannou, S. Rajput, J.-y. Sohn, K. Lee, J. D. Lee, and D. Papailiopoulos. Looped transformers as programmable computers. In *International Conference on Machine Learning (ICML)*, pages 11398–11442, 2023.

[13] S. McLeish, A. Bansal, A. Stein, N. Jain, J. Kirchenbauer, B. R. Bartoldson, B. Kailkhura, A. Bhatele, J. Geiping, A. Schwarzschild, and T. Goldstein. Transformers can do arithmetic with the right embeddings. In *International Conference on Machine Learning (ICML)*, 2024.

[14] A. Kazemnejad, I. Padhi, K. Natesan Ramamurthy, P. Das, and S. Reddy. The impact of positional encoding on length generalization in transformers. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

[15] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, E. Zettlemoyer, N. Cancedda, and T. Scialom. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

[16] A. Ruoss, G. Delétang, T. Genewein, J. Grau-Moya, R. Csordás, M. Bennani, S. Legg, and J. Veness. Randomized positional encodings boost length generalization of transformers. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

[17] Y. Zhou, U. Alon, X. Chen, X. Wang, R. Agarwal, and D. Zhou. Transformers can achieve length generalization but not robustly. *arXiv preprint arXiv:2402.09371*, 2024.

[18] S. Duan and Y. Shi. From interpolation to extrapolation: Complete length generalization for arithmetic transformers. *arXiv preprint arXiv:2310.11984*, 2023.

[19] Y. Fan, Y. Du, K. Ramchandran, and K. Lee. Looped transformers for length generalization. *arXiv preprint arXiv:2409.15647*, 2024.